

Basic Proposal — Line-by-Line Rationale

Companion to `finestst\_proposal\_basic.md`. For every block of the proposal, this doc answers two questions: why the line is there (what it is doing to the reviewer or guarding against) and how it works (the mechanism or evidence behind it). Internal document — never submitted.

Title

"Measuring and Explaining Geomorphic Change in the McMurdo Dry Valleys, Antarctica"

Why: the title is a promise, and this one promises exactly two verbs — measure and explain. No "novel," no "framework," no method name. Reviewers read dozens of titles; plain verbs stand out and are impossible to overclaim. **How:** "Geomorphic Change" (not "Stream-Channel Change") matches the O2 rescope — the project measures the whole valley floor. "McMurdo Dry Valleys" names the place specifically enough that a panelist who knows the region is instantly oriented.

Summary block

"The McMurdo Dry Valleys hold the only streams in Antarctica that flow over open ground."

Why: the first sentence must establish stakes in one breath for a reviewer who may know nothing about the region. "Only streams in Antarctica" is a superlative that is *true*, cheap to verify, and does the entire why-should-anyone-care job. **How:** it also quietly scopes the project — streams are the anchor even though O2 widens the view.

"A 2026 dissertation (Barlow) mapped two decades of that change but stopped short of explaining it — and measured it only inside the stream channels."

Why: names the prior work immediately and honestly. Reviewers distrust proposals that bury their intellectual parentage; leading with it converts a vulnerability ("isn't this just her work?") into the setup for the gap. **How:** the two clauses are the two objectives in embryo: "stopped short of explaining" → O1; "only inside the stream channels" → O2. Every later section unpacks this sentence.

"This document is a plan. No results are claimed here..."

Why: the plan-only rule, stated in the proposal's own voice. It preempts the reviewer question "where are the preliminary results?" by making their absence a deliberate posture rather than an omission. **How:** it reframes what the reviewer should score: the plan's design quality, not a portfolio of prior outputs.

§1 Background

"...largest ice-free region of Antarctica. For most of a century they were treated as the most stable landscape on Earth. That view is now in question."

Why: three sentences that set up the entire scientific tension: stability was the consensus (cosmogenic dating gives erosion rates of tens of cm per *million* years; Miocene ash deposits still sit undisturbed), and change against that baseline is therefore loud. A reviewer needs this contrast to feel why small changes matter here. **How:** "treated as" (not "is") keeps the claim historical and safe; "in question" states the current situation without overclaiming that stability is dead.

"The valleys are energy-limited: frozen water is everywhere, but there is barely enough summer heat to melt any of it."

Why: this single definition carries the whole causal argument of O1. If heat is the scarce input, then heat — not water supply — should control the response. H1 is this sentence turned into a testable statement. **How:** defining "energy-limited" at first use follows the rule that a proposal must parse on first read by a non-specialist panelist.

"What already exists" (three bullets: terrain-only U-Net, REMA validation, three-epoch change maps)

Why: the fair-credit ledger. Each bullet is something the proposal does NOT need to invent, which shrinks the perceived risk of the plan. Reviewers score feasibility; showing that the hard foundational pieces are published work makes the plan look buildable without claiming any of it as ours. **How:** the three bullets deliberately mirror Barlow's three methodological pillars, in the order the pipeline runs: detect → validate the ruler → difference.

"What does not exist... it does not test why... and its analysis is masked to the stream channels, so everything the elevation differencing measures outside them... was discarded unexamined."

Why: the gap statement is the proposal's reason to exist, so it names both gaps in parallel structure. "Discarded unexamined" is the strongest honest phrasing: the measurement of the whole surface already happens inside a DoD; Barlow's channel mask throws most of it away. That framing makes O2 sound like picking up dropped data, not inventing a new instrument — which is exactly what it is. **How:** "named in the dissertation as future work" (for drivers) shows the gap is real by the prior author's own account, not manufactured by us.

"Relevance to NASA..."

Why: Relevance to SMD is a *scored criterion* in FINESST review; a proposal without an explicit relevance statement forfeits points a paragraph can win. **How:** four arguments in rising order of reach: (1) the 2001 baseline literally *is* NASA data (ATM LiDAR) being revived; (2) the methods transfer to NASA missions (ICESat-2, NISAR elevation records); (3) the MDVs are NASA's canonical planetary-analog terrain; (4) the FI development itself matches the Division's open-science goals. If a reviewer rejects one leg, three remain.

§2 Objectives

"O1 — Attribution. Determine which climate drivers control each stream's rate of geomorphic change."

Why: O1 is first because it is the headline science and the reason the project is a *science* proposal rather than a mapping exercise. Everything else in the proposal either feeds it (O2 gives it more change types to explain, O3 makes its inputs trustworthy) or supports it. **How:** the driver list (PDD, sunlight, discharge, glacier mass balance, thaw depth) is exactly the list of datasets in §4 — every named driver is backed by a named, free data source. No driver is promised that can't be built.

"Hypothesis: cumulative melt energy predicts sediment movement better than water volume alone. This is testable and falsifiable — if discharge alone predicts just as well, the hypothesis fails and that is a publishable answer too."

Why: review panels reward hypotheses that can lose. Stating the failure condition in the objective itself signals scientific maturity and removes the "what if you're wrong?" attack — being wrong is a listed outcome. **How:** the energy-vs-water contrast is not arbitrary: in an energy-limited system (§1), energy *should* win. The hypothesis is the background section cashed out.

"O2 — Landscape-wide geomorphic change.... classify each patch of significant change by the process responsible: channel shift, thaw-driven subsidence (thermokarst), slope movement, fan growth, lake-margin change."

Why: this is the project's scope claim — geomorphology broadly, not channels only. The five named process types make "landscape-wide" concrete instead of hand-wavy, and each type is a real MDV phenomenon (thermokarst and lake-level rise are documented in the region; lake margins move meters because closed-basin levels integrate melt). **How:** classification is the bridge to O1: each process type gets its own driver test. That is what H2 means by "distinct driver fingerprints" — channels should follow melt energy and water, thermokarst

should follow thaw depth. If the types *share* one driver, the sentence "that is also an answer" makes the null explicit and publishable. Note also what the classes quietly handle: lake-margin change is mostly a *water-level* signal, not sediment transport — giving it its own class keeps it from contaminating the geomorphic classes.

"One practical note on satellite data... Measuring and correcting that disagreement is part of the method (§3), not an objective."

Why: this paragraph is the tombstone of the old O2 (cross-sensor generalization). The sensor problem is real and a reviewer will think of it, so the proposal must show it is handled — but handled as plumbing, not headline. Demoting it explicitly prevents a reviewer from mistaking the project for a remote-sensing methods exercise.

How: one sentence of stakes (the record only continues on REMA), one sentence of placement (it's in §3). Nothing more — proportionality is the message.

"O3 — Calibrated uncertainty.... on ground that did not change, a 95%-confidence threshold should be exceeded by roughly 5% of pixels — no more, no less."

Why: O3 is what makes O2 credible. Outside the channels there is no prior on where change should appear, so the only defense against a map full of noise is a threshold that provably behaves. It also produces the most reusable deliverable (an uncertainty-layer recipe any DEM-differencing study can copy). **How:** the "no more, no less" calibration test is strict on purpose: exceeding 5% means the threshold lies (false confidence); far under 5% means it wastes real signal (too conservative). Verifiable calibration is a claim very few change-detection studies make — it's the proposal's quality signature.

"Together the three objectives turn a one-time survey into a continuing, self-checking monitoring capability with an explanatory model behind it."

Why: the closing line answers "so what do we have at the end?" in one sentence — the difference between three projects stapled together and one program.

§3 Planned approach

O1 step 1: "...keep only changes larger than the detection floor, sum inside the stream's channel outline, normalize by channel area and by years elapsed."

Why: each clause kills a specific artifact. Thresholding kills noise counted as change; area-normalizing kills "bigger footprint = bigger number" (survey footprints differ between epochs); per-year rates make epochs of different lengths comparable. **How:** stating the normalizations here — rather than discovering them in Year 1 — shows the FI already knows where this analysis usually goes wrong.

O1 step 3: "Fit two model families — hierarchical regression (interpretable) and random forest (flexible)... Validate only on streams held out of fitting. Agreement between the two families is the sanity check."

Why: two families because each covers the other's weakness: regression gives interpretable coefficients but assumes a functional form; a random forest fits anything but explains little. Agreement means the signal is in the data, not the model choice. **How:** held-out-stream validation is the honest version of "our model fits well" — with few streams, in-sample fit is nearly meaningless, and a reviewer who knows small-n statistics will look for exactly this sentence.

O2 steps 1–3 (threshold the whole floor → classify by shape and setting → per-type driver tests)

Why: the three steps show O2 is a pipeline, not a wish: detection is inherited from the existing DoD machinery (nothing new to invent), classification uses observable evidence (slope position, distance to channels and lakes, ice-cored terrain), and the output feeds directly into O1's already-built driver models. **How:** ordering matters —

thresholds come from O3, so O2 is explicitly *gated* by O3. The dependencies among objectives are stated, which is what makes the Year-2 placement in §6 credible.

"Sensor continuity (method, Years 1–2)..."

Why/How: same demotion logic as §2's note — present so the reviewer sees it is scheduled and scoped, small so it can't be mistaken for the point.

"Methods note. All elevation differencing uses robust statistics (NMAD...) because elevation errors in this terrain are known to be heavy-tailed..."

Why: one technical commitment stated up front. Elevation error in this terrain has a few large blunders (heavy tails); a standard deviation would let those blunders inflate the noise estimate and hide real change. **How:** "known to be heavy-tailed" leans on the published error analysis (Höhle & Höhle; Barlow's own error model) without claiming any measurement of ours — consistent with plan-only.

§4 What the project needs

The data table, with an "Availability" column reading "public, free" — and one row reading "author-gated" — must be requested."

Why: the table exists to let a reviewer verify in ten seconds that the project cannot be data-blocked. The column says "availability," not "status" or "in hand" — the plan-only rule extends to not implying anything has been fetched. **How:** putting the single gated item *in the same table* rather than hiding it in prose is deliberate: the honesty is visible at the same glance as the mitigation pointer.

"Labels.... If that request fails, a comparable set will be re-digitized by hand. This costs weeks, not months, and is budgeted in Year 1."

Why: the label dependency is the proposal's only real external risk, so it gets its own paragraph with a fallback that depends on nobody. "Weeks, not months" sizes the fallback so the reviewer doesn't have to guess whether it wrecks the timeline. **How:** the claim is credible because hand-labeling terrain features is the FI's documented prior experience (Research Readiness Statement carries the detail).

"Computing. A single workstation GPU is sufficient... No supercomputer allocation is required."

Why: removes an entire category of reviewer doubt (infrastructure risk) in three sentences. Modest needs also signal that the budget will be spent on the student, not on hardware.

"Skills and mentoring. The FI... is new to Antarctic hydrology and to hierarchical statistical modeling — a genuine gap, stated as one."

Why: FINESST's third scored criterion is Research Readiness, and its purpose is training. A proposal with no gaps reads as either dishonest or as not needing the fellowship. Naming the two real gaps — and pointing to where the closing plan lives (RRS, Mentoring Plan) — turns a weakness into program fit. **How:** the gaps chosen are real but bounded: both are closable by coursework and advisor mentorship, neither blocks Year-1 work (which is pipeline construction, the FI's demonstrated strength).

"Community contact. The dissertation author has been contacted and is responsive. Continued cooperation is helpful but not load-bearing..."

Why: answers two opposite reviewer worries at once — "do they have access to the prior group?" (yes, contact exists) and "does the plan collapse without that goodwill?" (no, everything runs from public data). **How:** "not load-bearing" is the operative phrase; it is verifiable against §4's table, where every required dataset is public.

§5 Honest assessment

The section exists at all.

Why: most proposals bury risk in two mitigative sentences. A section titled "Honest assessment" that leads with "What could go wrong, and what happens then" is a credibility play: it does the adversarial review *for* the panel, which leaves them less to write and signals the FI can be trusted with three unsupervised years. **How:** every risk follows the same grammar — failure mode, consequence, what gets delivered anyway. No risk is listed without a landing zone.

Risk 1: weak attribution signal → "a rigorous null result with calibrated uncertainty... The risk is to impact, not to feasibility."

Why/How: separates the two things reviewers conflate: whether the project can fail to *finish* (it can't — the measurement gets made regardless) versus fail to *excite* (possible, admitted). Feasibility risk is the killer in review; impact risk is survivable and honest.

Risk 2: gauges patchy after 2015 → "...the discharge side of the hypothesis test is weakest in the most recent period, and the proposal accepts that."

Why/How: this risk is also an argument *for* the design — modeled melt energy (continuous everywhere) is O1's independent variable precisely because gauges decay. The last clause ("accepts that") is there because pretending reanalysis fully replaces gauges would be overclaiming.

Risk 3: REMA coverage uneven → anchor on Taylor Valley; scope conclusions.

Why/How: pre-commits the fallback geography so a reviewer can't discover it as a gotcha. Taylor Valley genuinely has the best three-epoch coverage; claims degrade gracefully from "acceleration" (3 epochs) to "change" (2 epochs) elsewhere.

Risk 4: label request declined → re-digitize.

Why/How: closes the loop opened in §4. Redundant on purpose — risk sections get read in isolation.

Risk 5: outside-channel change may be too small to detect → "O2 narrows back to in-channel change and the landscape-wide result is reported as a calibrated null — which is itself a quantitative statement about how spatially confined active change currently is."

Why: the newest risk, created by the O2 rescope, and the most important to state: the whole-landscape ambition could genuinely find mostly noise-floor. The mitigation shows the failure mode converts into a *finding* — "active change is confined to X% of the surface at our detection limits" is a real, citable number. **How:** this only works because of O3 — a calibrated null means something; an uncalibrated null is just a shrug. The risks section quietly demonstrates the objectives' interdependence.

"What this proposal does not claim."

Why: the plan-only posture restated as a boundary. It also protects the FI in any later interview — no sentence in the document asserts work performed.

"Why it is worth doing anyway."

Why: an honest-assessment section that ends on risk would leave the reviewer in a down-beat. This paragraph is the recovery: the data exist, are free, and the missing ingredients (drivers, whole-landscape picture, error bars) are "exactly the size of a three-year graduate project" — which reframes the fellowship as the natural container for the work.

§6 Timeline

Why the shape: Year 1 = build + first science (attribution on Taylor Valley); Year 2 = the new-scope science (landscape-wide classification), *after* O3 exists to gate it; Year 3 = synthesis, prediction, and product release. Each year ends in a named deliverable because reviewers check whether a timeline produces papers or just activity. **How:** the dependencies run downhill — nothing in Year 2 needs anything that isn't a Year-1 deliverable. "Secure or rebuild training labels" sits in Year 1 because it's the longest-lead external dependency.

§7 Data management and open science

Why: a compressed echo of the separate 2-page OSDMP (which is the binding document). It exists in the S/T/M so the open-science posture is visible even to a reviewer who reads nothing but the six pages. **How:** the sentence structure mirrors NASA's expectations: what is produced, where it is released, what is deliberately not archived (heavy regenerable rasters), and what metadata every product carries.

References

Why these seven: [1] the dissertation (the foundation — citing it generously is both honest and strategic); [2] the peer-reviewed proof of concept (shows the foundation survived review, not just a committee); [3] Höhle & Höhle (the NMAD methodology the error handling rests on); [4] REMA (the dataset the future of the record depends on); [5] U-Net (the architecture's origin); [6] the LTER data catalog (the driver data's provenance); [7] ERA5 (the reanalysis gap-filler). Every reference is load-bearing — each one backs a specific claim in §1–§3, and nothing is cited for decoration.

Maintained alongside `finestst\_proposal\_basic.md` — when a proposal line changes, its entry here changes in the same pass.